



Machine learning-assisted structural parameters screening of Sb@C composites for high cycle capacity in sodium-ion battery

Yufei Wang, Haixin Zhou, Kuo Wang, Xin Liu, Youming Peng, Xueping Wu, LingJiao Chen, Jiaojiao Liang^{*}, Di Huang^{*}

College of Railway Transportation, Hunan University of Technology, Zhuzhou 412008, China

ARTICLE INFO

Keywords:

Sodium-ion batteries
Machine learning
Sb@C electrode
Structural parameter
High cycle capacity

ABSTRACT

Antimony@carbon (Sb@C) with high capacity and low reaction potential have been the focus of research on sodium storage electrodes. However, it is challenging to obtain the precisely correlation of electrode structure parameters on battery capacity in experimental research. Machine learning (ML) is a kind of artificial intelligence and can predict the quantitative relationship between material structural and cycle capacity of electrodes by building models. Hence, in this work the 6 key structural characteristics parameter of Sb@C electrode are investigated for sodium-ion batteries with cycle performance by 4 kinds of algorithms based on ML, and Ada-boost algorithm embody the best performance with R^2 values of 0.91 and 0.83 on the test set and training set with 10-fold cross-validation, respectively. Furthermore, it is obtained the antimony content (AC) of Sb@C composite is the most critical of structural parameters affect the remaining capacity of cycle performance through feature screening and SHAP analysis, due to the sodium storage mechanism that the cycle capacity of Sb@C electrode are directly affected the high capacity of Sb. Meanwhile, it is accurately predicted the 0.65–0.75 of AC at smaller-size of antimony size (AS) and higher or lower of the intensity ratio for Raman peak (RP) with Sb@C electrode is obtained the high cycle capacity with exhaustion method. It corresponds that the stabilized structure of the carbon material effectively buffers the volume expansion caused by the addition of appropriate AC to reduce the capacity fading. And the higher or lower of RP value improves the electrical conductivity of the carbon and facilitates the electron transport. More importantly, the small-sized AS provide more active sites, which contribute to the storage and electrochemical reaction activity of sodium ions, thus further enhancing the cycle remaining capacity. In addition, the strong generalization ability of built model is validated by other series of metal@carbon composites within less 15 % of loss rate. The effective method and new direction with ML-screening are obtained to explore the structural parameter of electrode to predict battery cycle capacity for providing a comprehensive effective solution.

1. Introduction

Sodium-ion batteries (SIBs) have attracted a lot of attention because of low price and abundant sodium resources, which are expected to be used in low-speed electric cars, energy storage systems and portable electronics. As we all know, it is of great research significance for the rational structure of electrode materials with high capacity to improve the cycle performance of SIBs [1–3].

Antimony (Sb) has been widely studied as hot spot electrode materials in SIBs due to the high theoretical capacity of 660 mAh/g and good electrical conductivity. However, the larger volume expansion and contraction of Sb lead to electrode pulverization and structural collapse

in the charge-discharge process, thus affecting the cycle performance and specific capacity in SIBs[4,5]. Nanosizing Sb can better alleviate the stress concentration caused by the volume change to reduce the rupture and pulverization of electrode. Alternatively, Sb nanomaterials are composited with carbon to form Sb@C, in which the carbon material can provide the better electrical conductivity while acting as a buffer against volume expansion[6]. Many researchers have carried out different structural regulations of Sb@C composites to improve the sodium storage properties. For example, Huang et al., reported 70 wt% Sb in Sb@C by solvent method to form composites for SIBs, and the reversible capacity of 495.2 mAh/g after 100 cycles at a current density of 125 mAh/g[7]. Subsequently, Nguyen et al. investigated Sb@C

^{*} Corresponding authors.

E-mail addresses: liangjiaojiao@hut.edu.cn (J. Liang), dihuang@hut.edu.cn (D. Huang).

<https://doi.org/10.1016/j.jalcom.2025.179424>

Received 11 November 2024; Received in revised form 16 February 2025; Accepted 22 February 2025

Available online 24 February 2025

0925-8388/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

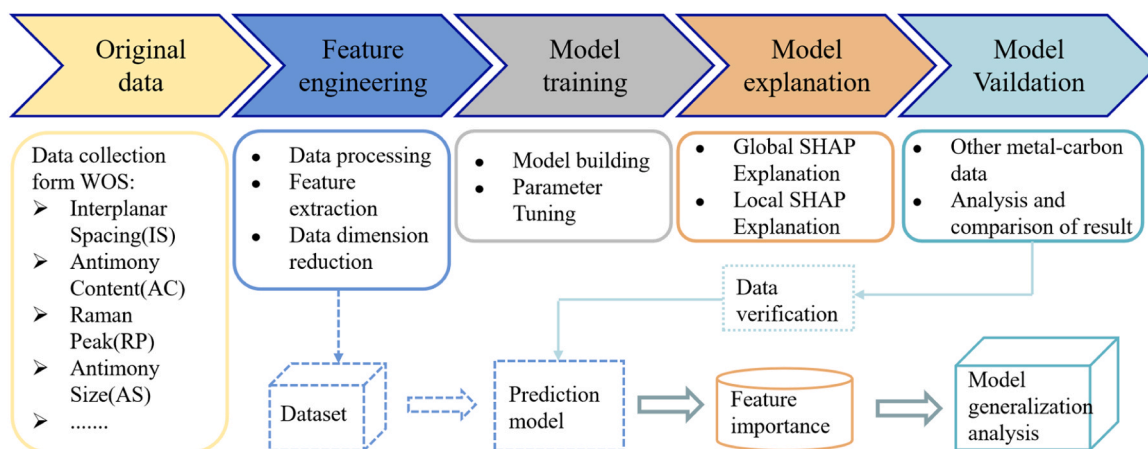


Fig. 1. The ML workflow for screening structural parameters of Sb@C electrode with high cycle remaining capacity for SIBs.

composites with Sb content of 79.19 wt%, 68.43 wt% and 62.70 wt% with corresponding reversible capacities of 571.2, 622.5 and 621.0 mAh/g after 100 cycles, respectively [8]. In addition, Liu et al. prepared porous Sb@C nanotube composites obtained the degree of crystallinity of 2.90, 2.23, and 3.21, respectively, which exhibited

reversible capacities of 290, 462, and 382 mAh/g after 100 cycles at a current of 100 mA/g [9]. Luo et al. discussed the Sb@C composites in which Raman peak intensity ratios of 3.12, 2.68 and 2.12 corresponded to reversible capacities of 587.4, 467.1 and 531.1 mAh/g after 100 cycles, respectively [10]. All show that the Sb content, degree of

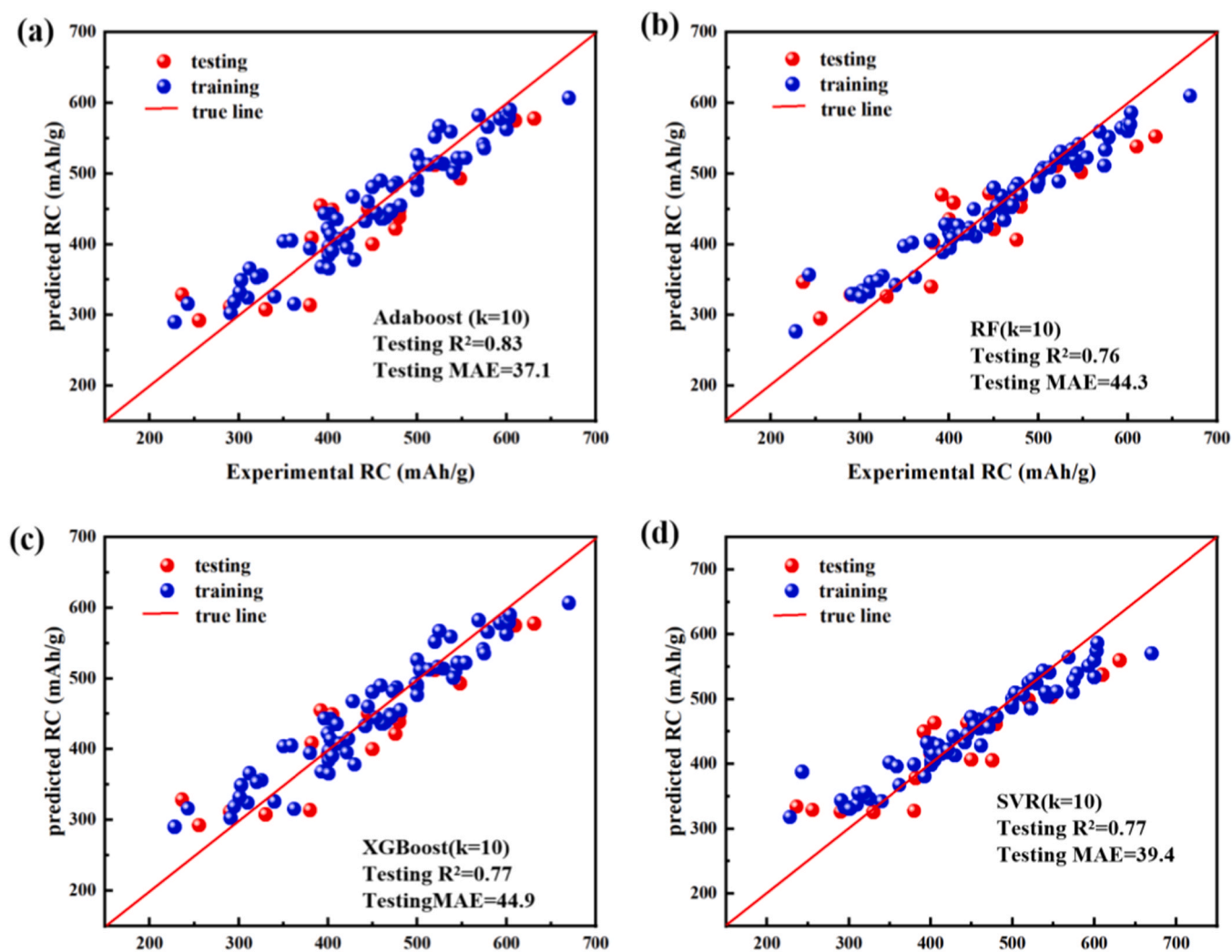


Fig. 2. The comparison between predicted values and experimental values of remaining capacity for Sb@C composite in SIBs (a) Adaboost model, (b) RF model, (c) XGBoost model (d) SVR model (In the figure, blue points represent the training set, red points represent the test set, and the red line represents the reference line where the predicted value is equal to the experimental value).

crystallinity and other structural parameters of Sb@C electrodes have the great influence on the cycle capacity of SIBs. However, the experimental research conclusions of a single variable cannot precise and comprehensive reflect the relationship between the structural parameters of Sb@C and the cycle capacity in SIBs.

Machine Learning (ML) is the core technology of artificial intelligence research, which is now gradually and widely used in the research design of electrode [11]. For example, Liu et al. used ML establish a suitable model to efficiently screen the effect of microstructural parameters of non-graphitic carbon materials based on sodium storage mechanism and performance through Web of Science databases[12]. As well as Isaiah A. et al. trained the ML models to predict the average voltage and volume changes of metal-ion battery electrode materials during charging and discharging based on the materials project database [13]. Subsequently, they successfully developed a tool and proposed nearly 5000 candidate electrode materials. Moreover, our group has predicted novel photoelectric materials to enhance power conversion efficiency by ML[14,15]. All above works show that ML is a highly efficient and adaptable method for designing and screening the structural of materials, providing an important way for the performance prediction.

Here, the structure parameters of Sb@C materials have been explored with ML to establish between structural parameters and cycle performance for SIBs with high remaining capacity in this work. The ML workflow for screening the structural parameters of Sb@C electrode with high cycle remaining capacity by ML for sodium-ion batteries as shown in Fig. 1. Firstly, 90 data points of original experimental research as displayed in Table S5 about the sodium storage of Sb@C composites with six key structural feature as inputs and cycle capacity as output were collected and extracted from the Web of Science in recent years. Then, the model was trained to predict cycle remaining capacity after normalization using ML such as Random Forest (RF), eXtreme Gradient Boosting(XGBoost), Adaptive Boosting (Adaboost) and Support Vector Regression (SVR). The Adaboost model exhibited the highest prediction accuracy with analysis by ML performance metrics. Meanwhile, the explainable analysis was also carried out by elucidating the effectiveness of the model through Shapley values, and the adoption of exhaustive method with constructing artificial data of the built model was applied to predict relatively precise range of structural characteristics, including AC, RP and AS, for the high cycle remaining capacity. Finally, the experimental data of sodium storage with other metal@carbon composite, liking Ge@C, Sn@C, Bi@C and Fe@C composites, were used to validate the generalization of the model.

2. Results and discussion

ML algorithms including Adaboost, SVR, XGBoost and RF were used to construct the training and testing models to establish and evaluate the mapping relationship between structural characteristics and cycle performance with high capacity in Sb@C composites, and the structural parameters as the input features and the remaining capacity of 100 cycles as the output feature were shown in Table S1, and the training results of the models were displayed in Fig. S1 without IC as the input features. Although Adaboost had better result compared with other models algorithms, the test set R^2 was 0.57 and the mean square error MSE was 54.2, which was relatively low. Because the R^2 value was used to evaluate and quantify the goodness of fit of the ML model and the relationship between the independent and dependent variables, and it was an important indicator in regression analysis[16], if the closer the R^2 value is to 1 with the higher the prediction accuracy of cycle remaining capacity. To improve the accuracy of the prediction model, the initial capacity (IC) was taken into account containing other factors, such as electrolyte type, synthesis method and other structure of material, etc. to affect remaining capacity of cycle performance and introduced as an input feature using the method of CEP.

The training and testing results of the ML algorithm using 10-fold

Table 1

Performance evaluation indicators of four models under 10-fold cross validation.

Model	Dat Set	R^2	RMSE	MAE
Adaboost	Training set	0.91	29.8	25.6
	Testing set	0.83	43.5	37.1
RF	Training set	0.86	37.3	26
	Testing set	0.76	51.8	44.1
XGBoost	Training set	0.87	34.2	26.1
	Testing set	0.77	50.9	43.2
SVR	Training set	0.87	41.3	27.2
	Testing set	0.77	44.3	38.3



Fig. 3. The correlation matrix with the key structural characteristics of Sb@C, initial capacity (IC), and cycle remaining capacity (RC).

cross-validation after adding IC were shown in Fig. 2 and Table 1. The R^2 values of the test sets and training sets for AdaBoost, RF, XGBoost and SVR were 0.83 and 0.91, 0.76 and 0.86, 0.77 and 0.87, 0.77 and 0.87, respectively. And the introduction of the initial capacity can contribute to improve the predicted performance evaluation metrics of all models which were greatly better optimized. At the same time, this also indicates the significant enhancement in the predictive capacity of the AdaBoost algorithm models for cycle remaining capacity. What's more, the RMSE (43.5) and MAE (37.1) values of the test sets for Adaboost models were lower than those of RF (RMSE was 51.8, MAE was 44.1), XGBoost (RMSE was 50.9, MAE was 43.2) and SVR (RMSE was 44.3, MAE was 38.3). This may be because RF, XGBoost, and SVR were prone to overfitting on small data sets or have difficulty capturing features effectively. The AdaBoost algorithm was simple and computationally efficient, and can adapt to data patterns in small datasets more quickly [17]. In addition, although both AdaBoost and XGBoost were boosting algorithms, XGBoost, as a more complex algorithm, usually can fully utilize its advantages on large-scale datasets, its complexity and computational overhead fail to bring about significant performance improvement in small datasets, but instead, it may increase the training time and heighten the risk of overfitting[18]. Therefore, the AdaBoost algorithm was selected as the best method to establish the quantitative relationship between the structural characteristics and the cycle remaining capacity on the small datasets. In addition, the 5-fold cross-validation of various algorithms were performed to prove the robustness of the model shown in Fig. S2 and Table S2. In contrast, 10-fold cross-validation of AdaBoost model is slightly better, with both RMSE and MAE on the test set lower than 5-fold cross-validation, and better generalization ability. And the two folds cross-validation of other algorithms also showed the similar results, which is reason that the higher percentage of training data could better capture the features,

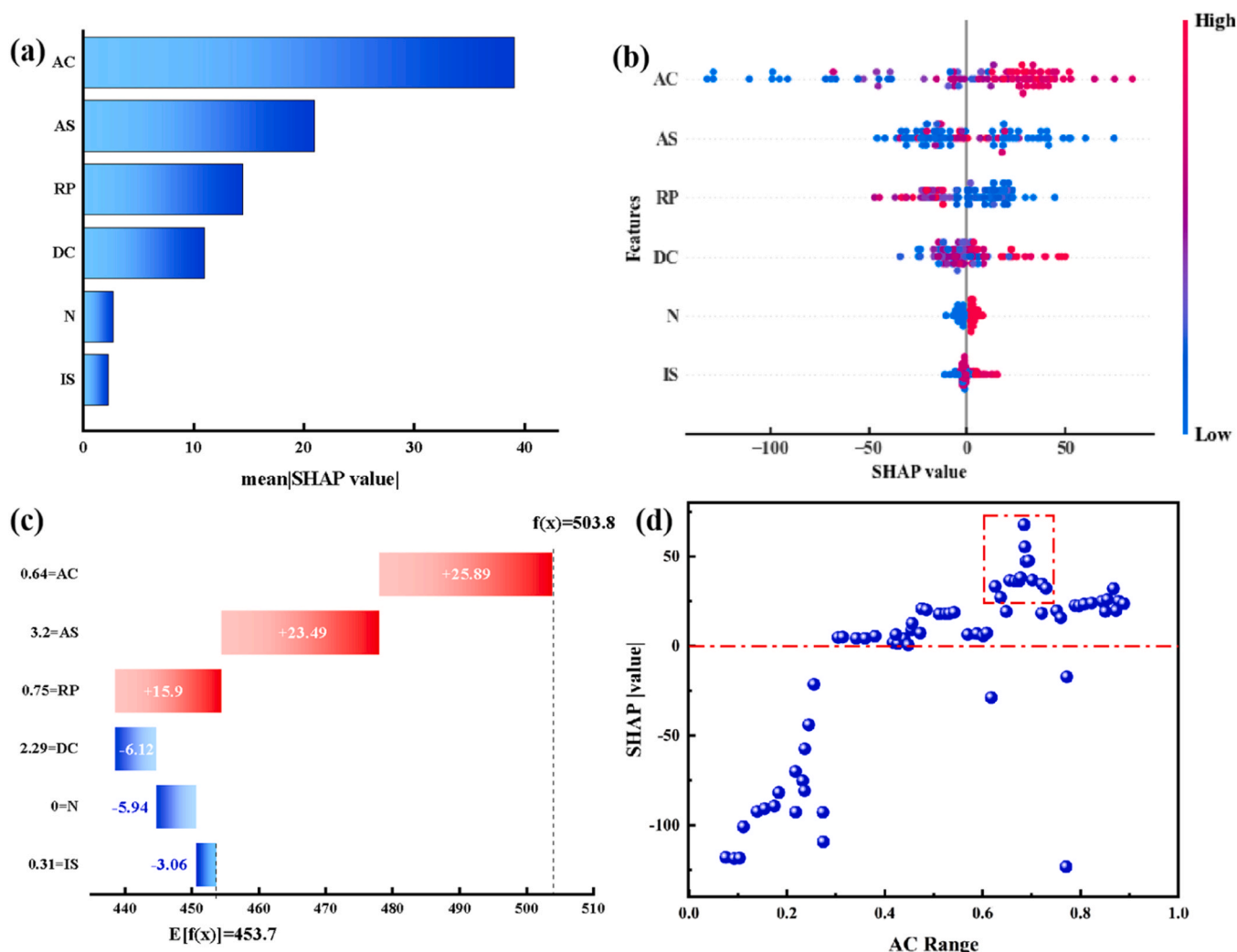


Fig. 4. (a) The SHAP plots of the Adaboost-RC model, (b) The impact of a certain feature on RC, (c) Waterfall plot of predicted cycle remaining capacity, (d) SHAP value of a single feature for Sb content (AC).

while more folds reduce the effect of uneven distribution of test data. Therefore, the different cross-validation methods are highly consistent and the performance of the test set was relatively stable, and it indicated that the model exhibits consistent robustness under different data splits without significant overfitting or underfitting. Therefore, the AdaBoost algorithm with 10-fold cross-validation was selected as the best method to establish the quantitative relationship between the structural characteristics and the cycle remaining capacity.

For analyzing the correlation strength between structural characteristics and cycle remaining capacity of Sb@C composites, the Pearson correlation coefficients (r) between each input feature and cycle remaining capacity were calculated and shown in Fig. 3, and it could be seen 0.55 of r between AC and RC, which suggested that the AC should be considered to optimize the cycle performance with high capacity, and Pham et.al[19] reported the antimony content was increased from 0.58 to 0.64 and the corresponding remaining capacity of cycle performance was increased from 498 mAh/g to 548 mAh/g. In addition, there was a significant correlation between the RP and AS with RC, and the Pearson correlation coefficients were correspond to 0.22 and 0.14, respectively. This suggested that the cycle remaining capacity was comparatively affected by RP and AS. Moreover, the Pearson correlation coefficient of 0.72 was obtained and shown a better statistical trend between the initial capacity and the output characteristic of cycle remaining capacity from the calculation, and it was indicated the cycle remaining capacity

were greatly affected by the initial capacity. Meanwhile, this was also a significant correlation ($r = 0.40$) between the IC and AC and it was consistent with the accordingly raised initial capacity with the increase of AC due to the high theoretical capacity of 660 mAh/g for Sb, and these are the main reasons why the IC was added as input feature by CEP.

To the quantitative analysis of the influence and contribution of the input structural characteristics to the cycle remaining capacity, the particular contribution value of every input feature to the output was calculated with the SHAP method based on the Adaboost model, as shown in Fig. 4. The average absolute of the SHAP value for every feature were represented the relevance of the major characteristics, and the more relevant the feature was the higher the SHAP value. Fig. 4a showed the SHAP plots for the Adaboost model of cycle remaining capacity and the effect of a certain feature on the RC, and Fig. 4b displayed that each sample was affected by each feature of the target variable, with each point representing an independent sample, and blue color indicated a negative correlation and red color indicated a positive correlation[20]. It was clear that AC was the most important input feature among all structural features and AC positively correlation affected the RC values in Sb@C composites for sodium-ion storage. These results were consistent with the sodium storage mechanism for the high-capacity of Sb@C electrode, which means the content of antimony directly affects the capacity of Sb@C composites since the theoretical

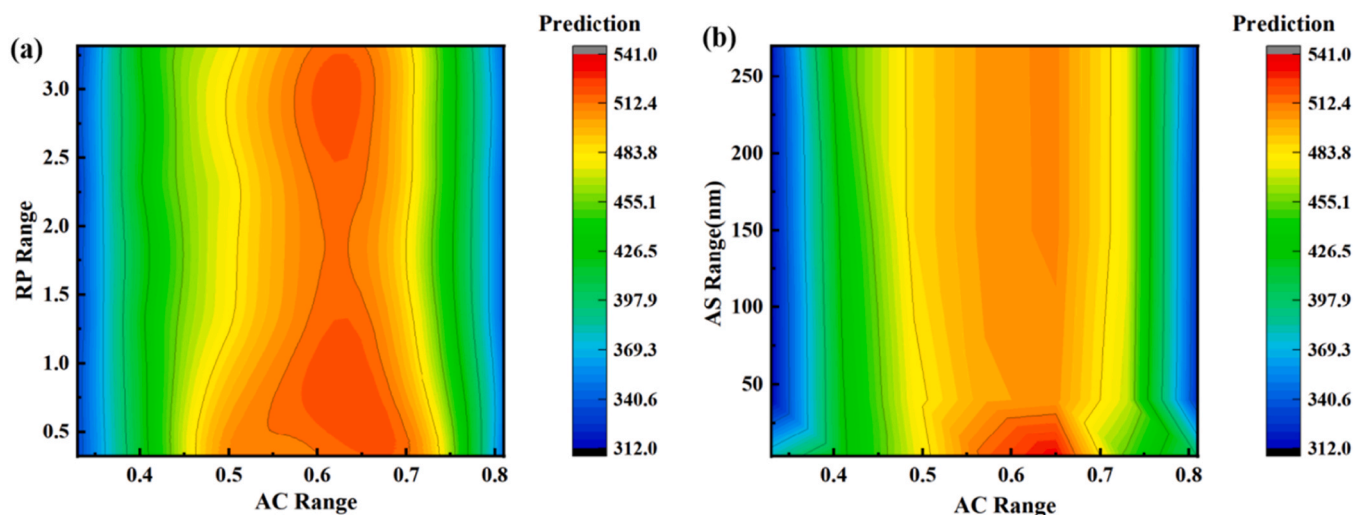


Fig. 5. The contour maps of change trends for predicted remaining capacity with structural parameters of (a) AC and RP, (b) AC and AS.

capacity of Sb was as high as 660 mAh/g. And the higher capacity for Sb@C electrode was provided with composition of antimony, and the stable structure of carbon material can buffer the volume expansion of Sb to avoid the attenuation of remaining capacity of cycle performance [21]. Meanwhile, the SHAP values of AS and RP showed the negative correlation with RC values, which indicated that the smaller the value of AS and RP, it was probable to be the higher of remaining capacity. The reason might be that the smaller RP possessed a higher degree of graphitization of the carbon material [22], which had more ordered structure and better conductivity, which contribute to the transport of electrons [23], in favor of improving the cycle performance and higher remaining capacity for SIBs. Similarly, the smaller-sized antimony particles had more specific surface area to provide more active sites in favor of storing sodium-ion and improving the electrochemical reaction activity to enhance the cycle remaining capacity of SIBs [24,25]. In addition, small-sized antimony particles were not prone to significant structural damage due to the easy dissipation of tension, and small-sized particles may be more suitable for maintaining the structural integrity of the material to reduce the capacity decay and improve the cycling remaining capacity of Sb@C electrode [26]. In addition, compared to another structure characteristics for DC, IC and N of Sb@C composites influencing the cycle performance, the change of AC can affect the sodium storage capacity of SIBs more directly, while AS and RP affect the cycling remaining capacity more indirectly through the internal structure. Therefore, the AC, AS and RP of structural parameters in Sb@C composites should be obtained more attention on the design of high-performance electrode materials.

To further analyze the local effects of individual structural parameters on cycle remaining capacity, the result of the local SHAP interpreter based on the Adaboost model was shown in Fig. 4c. The $E[f(x)]$ represented the average predicted value of the model over all samples, which was equivalent to the model's starting point of 453.70 mAh/g, when the contributions of particular structural characteristics were not taken into account with the cycle remaining capacity. When the structural characteristic of AC was 0.64, the predicted value of cycle remaining capacity was increased 25.89 mAh/g. Meanwhile, the AS for 3.20 nm and RP for 0.75 contributed to the corresponding predicted value for + 23.49 and + 15.9 mAh/g, respectively. Furthermore, 2.29 nm of the DC, nitrogen free and 0.31 nm of IS reduced the predicted values of 6.12, 5.94 and 3.06 mAh/g, respectively. The final result of cycle remaining capacity from the predicted model can be obtained as 503.8 mAh/g, which was superimposed the individual feature on this baseline value. Meanwhile, it was discovered that AC was the most important factor in the model's decision-making process. Therefore, the greatest impact of

the input feature for AC was chosen to do a single SHAP analysis as displayed in Fig. 4d, and the degree of contribution for AC was overall increasing trend to the cycle remaining capacity. When the AC was after increased to 0.4, the SHAP value was positive, indicating that this region of the AC has a certain positive contribution to the prediction results. The SHAP value increased steadily with the enhancement of AC, reaching a notably high range between 0.65 and 0.75. This indicates a positive contribution toward achieving the desired cycling remaining capacity.

To precisely guide the experimental preparation through the prediction model, the sets of artificially constructed structural data were constructed based on the existing value range of each structure-performance data [27] (Table S3), in which the IC was determined by the approximate change trend between AC with the 4-means clustering method to fit a curve (Fig. S3). And other values of the input feature to be collected in the partition segments. And it was performed ML modeling on the above complete structural property database. Firstly, the original 90 existing data were divided into training sets and test sets in a ratio of 8:2 to input into the model, and the suitable model was established through multiple iterations. Then, the prediction model was obtained from the artificially constructed data, which eliminated unqualified theoretical data and substituted into the above model. The key parameters of structural characteristics for AC, RP and AS were chosen to illustrate the laws of change for obtaining specific structures and properties. The contour maps of the simultaneously effect for predicted remaining capacity with structural parameters of AC and RP as displayed Fig. 5a, and the predicted values of high remaining capacity were concentrated in the range of RP at lesser than 1.25 and greater than 2.5, the AC in the range from 0.65 to 0.75. It revealed the higher capacity of the Sb@C composite was provided with the moderate amount of AC, due to the high theoretical capacity of Sb, and the lesser of RP had well-ordered structure and better electrical conductivity for the transport of sodium ions and obtaining the high cycle remaining capacity [22], which was a consistent with the above conclusion of the SHAP values. In addition, another high cycle remaining capacity region of the Sb@C composite was also exhibited, when the Sb@C composite has RP was greater with more defects and disordered structures of carbon material to increase the embedding sites for sodium ions [28–30], which was reasonable explanation that the best cycle remaining capacity can be obtained for Sb@C composite with an interval range of RP. Fig. 5b demonstrated the collaborative effect of AS and AC on the predicted cycling remaining capacity, where the expected high-value region of sodium storage capacity was concentrated in the range of about 3 nm to 25 nm for AS and also closing to 0.65 for AC, which corresponded to the

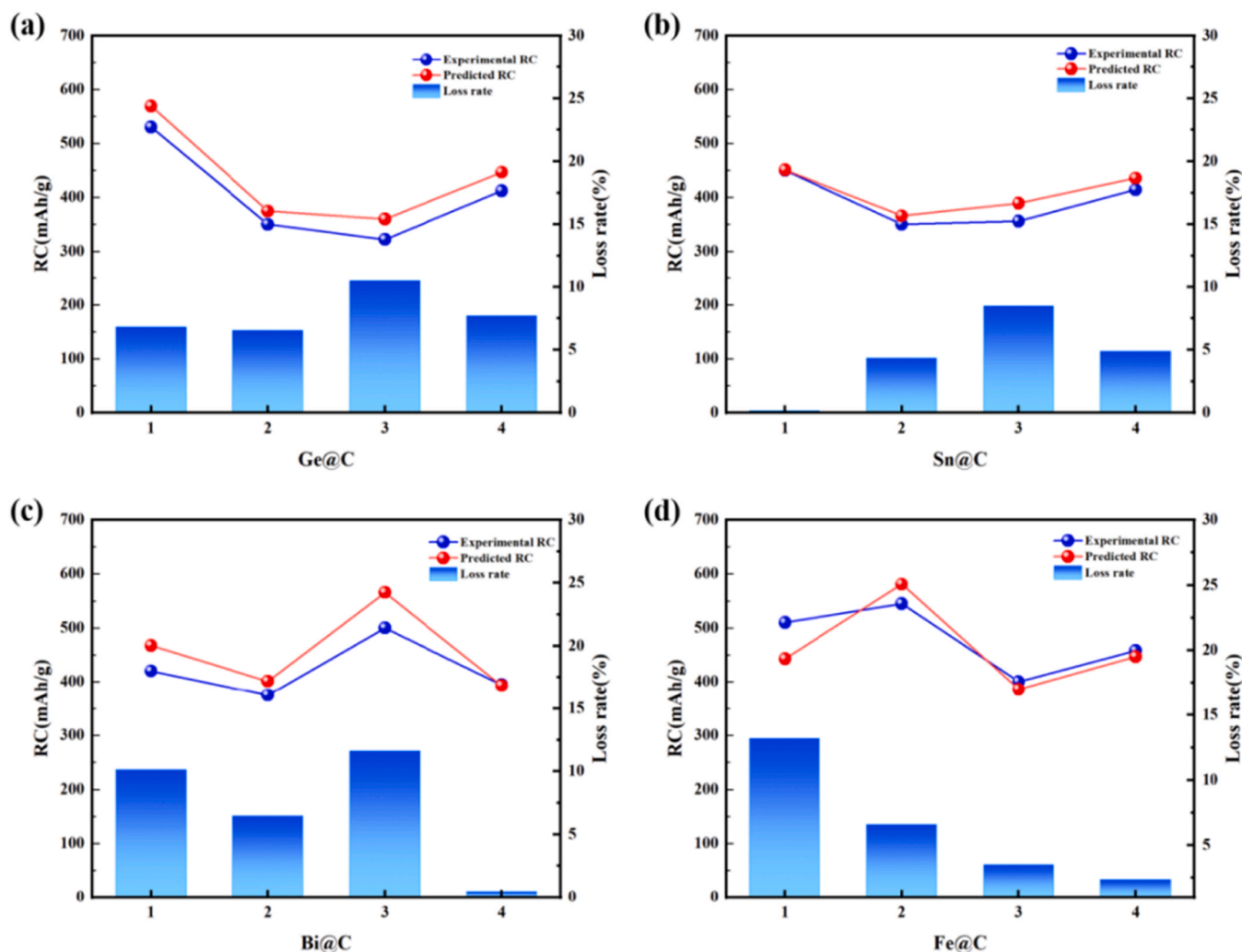


Fig. 6. The comparison of the experimental data and predicted data for (a) Ge@C, (b) Sn@C, (c) Bi@C and (d) Fe@C.

above analysis of the SHAP. This means the small-sized particle of Sb@C composite was better suited for preserving the structural integrity of the material to increase the cycle remaining capacity, but the larger size of antimony particles was readily triggered with mechanical stress and strain during the charging and discharging process, which causes the particle rupture and the pulverization of the electrode material, rapidly degrading the cycling capacity of sodium-ion battery[26].

In order to further verify the generalization ability and accuracy of the built model, the experimental results of other metal-carbon composites with the similar reaction mechanisms of Sb@C composite were used to validate the model for the cycle remaining capacity. Fig. 6 and Table S4 showed the predicted outcomes and loss rate in the sets of experimental data for the sodium storage property of Germanium-carbon (Ge@C) [31–34], Tin-carbon (Sn@C)[35–38], Bismuth-carbon (Bi@C) [39–42] and Iron-carbon (Fe@C) [43–46]. The loss rate of the model was computed in the $(|\text{Experimental RC} - \text{Predicted RC}|) / \text{Experimental RC}$. It was observed when the experimental cycling capacity of 322 mAh/g and predicted capacity of 359.6 mAh/g for the maximum loss rate of 10.47 % in Ge@C material, and other loss rate of was all less than 15 %. And there was a certain difference compared with the Sb@C material, which may be from the different theoretical capacities of other metals, thus affecting the cycling capacity of SIBs. Therefore, the established prediction model of cycle remaining capacity can be transferred to other metal composites and has strong generalization ability.

3. Conclusion and future

In summary, this work established the prediction model for the high remaining capacity of Sb@C electrode for sodium-ion storage based on the multiple types of ML algorithm, and it was shown that the Adaboost algorithm had the more accurate predictions compared with XGBoost, RF, and SVR, and the corresponding R^2 of the test set reaches 0.83, with an average absolute error of 37.1. Moreover, the consistent robustness of prediction model was obtained with under different fold cross-validation without significant overfitting or underfitting. Meanwhile, it quantitatively analyzed the relationship between the structural characteristics of Sb@C composite with the cycle remaining capacity through the established prediction model, and the better cycling capacity could be predicted from the range of Raman peak intensity ratio for 0.5–1 or 2.5–3, antimony size for 3–25 nm and antimony content for 0.65–0.75. In addition, the build model of cycle remaining capacity for Sb@C composite was verified by the method of data migration with the sodium storage properties of other metal@carbon composite, and the error rate of the model is less than 15 %. Our work establishes a database of structural properties of antimony@carbon (Sb@C) composites and analyzes the relationship between structural parameters and sodium storage performance using machine learning. An efficient methodology is proposed, which helps to correlate the material structure with high cycling capacity to design and screen electrode materials for high-performance sodium-ion batteries.

Meanwhile, in the future, the standardized processing criteria could

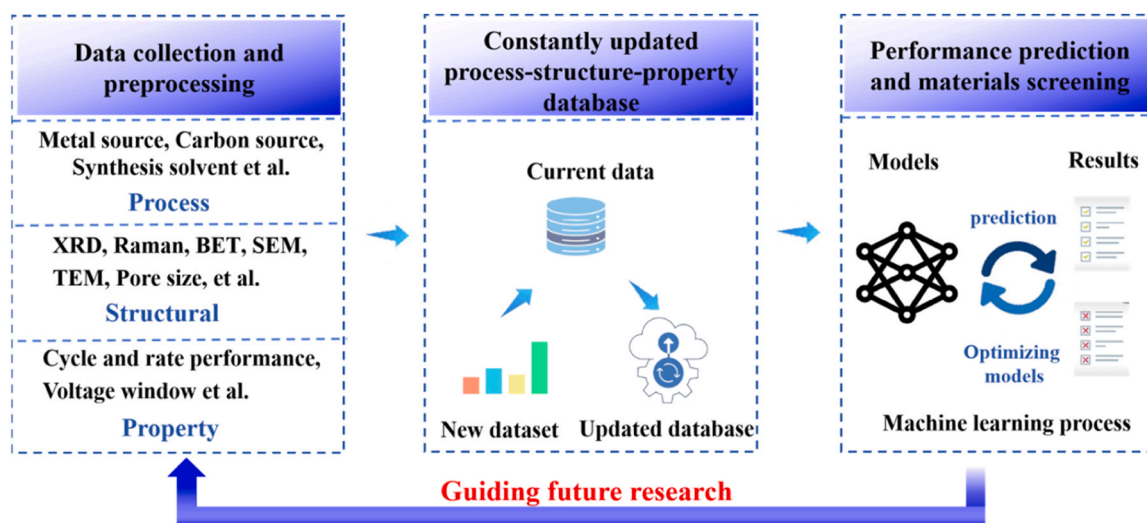


Fig. 7. The proposed process for the prediction of battery performance in the future base on ML.

be defined to develop structure-property database with using big data analysis and multidisciplinary cooperation for the effective design and preparation of high performance electrode material. The proposed process of data processing and modeling are displayed in Fig. 7. Firstly, the parameters of structural characteristics and energy storage performance for electrode material are obtained from the standardized processing and integrated into the structure-property database. Subsequently, the datasets are utilized to update the ML model from the technological side of the synthesis process and the structural characterization side for the electrode to directly screen and design high capacity materials. Furthermore, the process-property database can also be established with the preparation process of electrode materials, and the ML model can be trained to guide the design of experiments to minimize shortening the study period of high performance electrodes.

CRedit authorship contribution statement

Chen LingJiao: Supervision, Funding acquisition, Formal analysis. **Huang Di:** Writing – review & editing, Visualization, Methodology, Funding acquisition, Conceptualization. **Liang Jiaojiao:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Wang Yufei:** Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Zhou Haixin:** Writing – review & editing, Supervision, Software. **Wang Kuo:** Supervision, Project administration, Formal analysis. **Liu Xin:** Supervision, Resources, Investigation. **Peng Youming:** Supervision, Formal analysis. **Wu Xueping:** Validation, Supervision, Conceptualization.

Declaration of Competing Interest

All authors from the manuscript entitled as “Machine learning-assisted structural parameters screening of Sb@C composites for high cycle capacity in sodium-ion battery” declare that we have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the National Natural Science Foundation of China (No. 62106074), the Natural Science Foundation of Hunan Province (No. 2024JJ5127), the Education Department of Hunan Province (No. 22B0580 and 24A0421) and the Scientific research and innovation Foundation of Hunan University of Technology (No.

CX2317).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jallcom.2025.179424](https://doi.org/10.1016/j.jallcom.2025.179424).

Data Availability

Data will be made available on request.

References

- [1] L. Zhao, T. Zhang, W. Li, T. Li, L. Zhang, X. Zhang, Z. Wang, Engineering of sodium-ion batteries: opportunities and challenges, *Engineering* 24 (2023) 172–183.
- [2] Q. Xue, S. Li, Y. Zhao, P. Sheng, L. Xu, Z. Li, B. Zhang, H. Li, B. Wang, L. Yang, Y. Cao, Z. Chen, Novel Alkaline Sodium-Ion Battery capacitor based on active carbon||Na_{0.44}MnO₂ towards low cost, high-rate capability and long-term lifespan, *Acta Phys. Chim. Sin.* 40 (2024).
- [3] Z. Hu, Y. Niu, X. Rong, Y. Hu, Suppression of voltage decay through Ni³⁺ barrier in anionic-redox active cathode for Na-ion batteries, *Acta Phys. Chim. Sin.* 40 (2024).
- [4] N. Zhang, X. Chen, J. Xu, P. He, X. Ding, Hexagonal sb nanocrystals as high-capacity and long-cycle anode materials for sodium-ion batteries, *ACS Appl. Mater. Interfaces* 15 (2023) 26728–26736.
- [5] T. Perveen, M. Siddiq, N. Shahzad, R. Ihsan, A. Ahmad, M.I. Shahzad, Prospects in anode materials for sodium ion batteries - a review, *Renew. Sustain. Energy Rev.* 119 (2020).
- [6] C. Mi, Y. Qin, X. Huang, Y. Luo, Z. Zhang, C. Wang, Y. Shi, L. Yin, R. Wang, Galvanic replacement synthesis of graphene coupled amorphous antimony nanoparticles for high-performance sodium-ion capacitor, *Acta Phys. Chim. Sin.* 40 (2024).
- [7] K.-C. Huang, J.-Z. Guo, H.-H. Li, H.-H. Fan, D.-H. Liu, Y.-P. Zheng, W.-L. Li, X.-L. Wu, J.-P. Zhang, Layer-stacked Sb@graphene micro/nanocomposite with decent Na-storage, full-cell and low-temperature performances, *J. Alloy. Compd.* 731 (2018) 881–888.
- [8] A.-G. Nguyen, H.T.T. Le, R. Verma, D.-L. Vu, C.-J. Park, Boosting sodium-ion battery performance using an antimony nanoparticle self-embedded in a 3D nitrogen-doped carbon framework anode, *Chem. Eng. J.* 429 (2022).
- [9] H. Liu, Z. Wang, Z. Wu, S. Zhang, S. Ge, P. Guo, M. Hua, X. Lu, S. Wang, J. Zhang, Direct tuning of meso-/micro-porous structure of carbon nanofibers confining Sb nanocrystals for advanced sodium and potassium storage, *J. Alloy. Compd.* 833 (2020).
- [10] X. Luo, H. Tan, T. Ma, H. Wang, M. Lv, Z. Yu, C. Fu, X. Chang, S. Jin, Nitrogen doped porous carbon coated antimony as high performance anode material for sodium-ion batteries, *Nanotechnol* 32 (2021).
- [11] Z. Ahmed, K. Mohamed, S. Zeeshan, X. Dong, Artificial intelligence with multi-functional machine learning platform development for better healthcare and precision medicine, *Database* 2020 (2020).
- [12] X. Liu, T. Wang, T. Ji, H. Wang, H. Liu, J. Li, D. Chao, Using machine learning to screen non-graphite carbon materials based on Na-ion storage properties, *J. Mater. Chem. A* 10 (2022) 8031–8046.
- [13] I.A. Moses, R.P. Joshi, B. Ozdemir, N. Kumar, J. Eickholt, V. Barone, Machine learning screening of metal-ion battery electrode materials, *ACS Appl. Mater. Interfaces* 13 (2021) 53355–53362.

- [14] D. Huang, K. Wang, Z. Li, H. Zhou, X. Zhao, X. Peng, J. Wu, J. Liang, J. Meng, L. Zhao, A machine learning prediction model for quantitative analyzing the influence of non-radiative voltage loss on non-fullerene organic solar cells *Chem Eng J* 475 (2023) 145958.
- [15] K. Wang, J. Liang, Z. Li, H. Zhou, C. Nie, J. Deng, X. Zhao, X. Peng, Z. Chen, Z. Peng, D. Huang, H.S. Jang, J. Kong, Y. Zou, Design of experiments with the support of machine learning for process parameter optimization of all-small-molecule organic solar cells, *FlexMat* 1 (3) (2024) 234–247.
- [16] S. Sarkar, S.C. Peter, An overview on Sb-based intermetallics and alloys for sodium-ion batteries: trends, challenges and future prospects from material synthesis to battery performance, *J. Mater. Chem. A* 9 (2021) 5164–5196.
- [17] B. Dou, Z. Zhu, E. Merkurjev, L. Ke, L. Chen, J. Jiang, Y. Zhu, J. Liu, B. Zhang, G. W. Wei, Machine learning methods for small data challenges in molecular science, *Chem. Rev.* 123 (2023) 8736–8780.
- [18] Y. Cao, Q.-G. Miao, J.-C. Liu, L. Gao, Advance and prospects of adaboost algorithm, *Acta Autom. Sin.* 39 (2013) 745–758.
- [19] X.M. Pham, D.T. Ngo, H.T.T. Le, P.N. Didwal, R. Verma, C.W. Min, C.N. Park, C. J. Park, A self-encapsulated porous Sb-C nanocomposite anode with excellent Na-ion storage performance, *Nanoscale* 10 (2018) 19399–19408.
- [20] A.B. Parsa, A. Movahedi, H. Taghipour, S. Derrible, A.K. Mohammadian, Toward safer highways, application of XGBoost and SHAP for real-time accident detection and feature analysis, *Accid. Anal. Prev.* 136 (2020) 105405.
- [21] Y. Xiang, X. Hu, X. Zhong, W. Deng, G. Zou, H. Hou, X. Ji, Mechanism of fast storage of Li/Na in complex Sb-based hybrid system, *Adv. Funct. Mater.* 34 (2023).
- [22] L. Dashairya, D. Das, P. Saha, Electrophoretic deposition of antimony/reduced graphite oxide hybrid nanostructure: a stable anode for lithium-ion batteries, *Mater. Today Commun.* 24 (2020).
- [23] Y. Shen, Carbothermal synthesis of metal-functionalized nanostructures for energy and environmental applications, *J. Mater. Chem. A* 3 (2015) 13114–13188.
- [24] B. Chen, M. Liang, Q. Wu, S. Zhu, N. Zhao, C. He, Recent developments of antimony-based anodes for sodium- and potassium-ion batteries, *Trans. Tianjin Univ.* 28 (2021) 6–32.
- [25] A. Amardeep, R.C. Shende, P. Gandharapu, M.S. Wani, A. Mukhopadhyay, Faceted antimony particles with interiors reinforced with reduced graphene oxide as high-performance anode material for sodium-ion batteries, *ACS Appl. Mater. Interfaces* 14 (2022) 45296–45307.
- [26] Z. Liu, Y. Jiang, Q. Hu, S. Guo, L. Yu, Q. Li, Q. Liu, X. Hu, Safer lithium-ion batteries from the separator aspect: development and future perspectives, *Energy Environ. Mater.* 4 (2020) 336–362.
- [27] H. Gao, X. Guo, S. Wang, F. Zhang, H. Liu, G. Wang, Antimony-based nanomaterials for high-performance potassium-ion batteries, *EcoMat* 2 (2020).
- [28] I. Childres, L.A. Jauregui, W. Park, H. Cao, Y.P. Chen, Raman spectroscopy of graphene and related materials, *N. Dev. Photon Mater. Res.* 1 (2013) 1–20.
- [29] A. Eckmann, A. Felten, A. Mishchenko, L. Britnell, R. Krupke, K.S. Novoselov, C. Casiraghi, Probing the nature of defects in graphene by Raman spectroscopy, *Nano Lett.* 12 (2012) 3925–3930.
- [30] E. Pollak, G. Salitra, V. Baranchugov, In situ conductivity, impedance spectroscopy, and ex situ Raman spectra of amorphous silicon during the insertion/extraction of lithium, *J. Phys. Chem. C* 111 (2007) 11437–11444.
- [31] C. Liu, Y. Jiang, C. Meng, X. Liu, B. Li, S. Xia, Amorphous germanium nanomaterials as high-performance anode for lithium and sodium-ion batteries, *Adv. Mater. Technol.* 8 (2023).
- [32] Q. Li, Z. Zhang, S. Dong, C. Li, X. Ge, Z. Li, J. Ma, L. Yin, Ge nanoparticles encapsulated in interconnected hollow carbon boxes as anodes for sodium ion and lithium ion batteries with enhanced electrochemical performance, *Part. Part. Syst. Char.* 34 (2017).
- [33] N. Wen, S. Chen, J. Feng, K. Zhang, Z. Zhou, X. Li, Q. Fan, Q. Kuang, Y. Dong, Y. Zhao, In situ hydrothermal synthesis of double-carbon enhanced novel cobalt germanium hydroxide composites as promising anode material for sodium ion batteries, *Dalton Trans.* 50 (2021) 4288–4299.
- [34] M. Yanilmaz, G. Cihanbeyoglu, J. Kim, Centrifugally spun binder-free N, S-Doped Ge@PCNF anodes for Li-ion and Na-ion batteries, *ACS Omega* 8 (2023) 16987–16995.
- [35] Z. Liu, J. Li, J. Yang, Preparation of a novel gC₃N₄/Sn/N-doped carbon composite for sodium storage, *Chem. J. Chin. Univ.* (2021).
- [36] L. Luo, H. Qiao, W. Xu, D. Li, J. Zhu, C. Chen, Y. Lu, P. Zhu, X. Zhang, Q. Wei, Tin nanoparticles embedded in ordered mesoporous carbon as high-performance anode for sodium-ion batteries, *J. Solid State Electrochem.* 21 (2017) 1385–1395.
- [37] H. Ying, S. Zhang, Z. Meng, Z. Sun, W.-Q. Han, Ultrasmall Sn nanodots embedded inside N-doped carbon microcages as high-performance lithium and sodium ion battery anodes, *J. Mater. Chem. A* 5 (2017) 8334–8342.
- [38] Y. Li, C. Ou, J. Zhu, Z. Liu, J. Yu, W. Li, H. Zhang, Q. Zhang, Z. Guo, Ultrahigh and durable volumetric lithium/sodium storage enabled by a highly dense graphene-encapsulated nitrogen-doped carbon@Sn compact monolith, *Nano Lett.* 20 (2020) 2034–2046.
- [39] H. Kim, D. Kim, Y. Lee, D. Byun, H.-S. Kim, W. Choi, Synthesis of Bi₂S₃/C yolk-shell composite based on sulfur impregnation for efficient sodium storage, *Chem. Eng. J.* 383 (2020).
- [40] Y. Li, X. Zhong, X. Wu, M. Li, W. Zhang, D. Wang, Bi/C nanosheet microspheres with an open pore structure as anodes for sodium ion batteries with high capacity, excellent rate performance and long cycle life, *J. Mater. Chem. A* 9 (2021) 22364–22372.
- [41] J. Zhu, J. Wang, G. Li, L. Huang, M. Cao, Y. Wu, Heterogeneous structured pomegranate-like Bi@C nanospheres for high-performance sodium storage, *J. Mater. Chem. A* 8 (2020) 25746–25755.
- [42] J. Xiang, Z. Liu, T. Song, Bi@C nanoplates derived from (BiO)₂CO₃ as an enhanced electrode material for Lithium/Sodium-Ion batteries, *ChemistrySelect* 3 (2018) 8973–8979.
- [43] L. Yin, Y. Pan, M. Li, Y. Zhao, S. Luo, Facile and scalable synthesis of α -Fe₂O₃/ γ -Fe₂O₃/Fe/C nanocomposite as advanced anode materials for lithium/sodium ion batteries, *Nanotechnol* 31 (2020).
- [44] H. Qi, C. Zhao, J. Huang, C. He, L. Tang, W. Deng, Metastable FeCN₂@nitrogen-doped carbon with high pseudocapacitance as an anode material for sodium ion batteries, *Nanoscale* 14 (2022) 780–789.
- [45] S. Zhang, J. Mi, H. Zhao, W. Ma, L. Dang, L. Yue, Electrospun N-doped carbon nanofibers confined Fe_{1-x}S composite as superior anode material for sodium-ion battery, *J. Alloy. Compd.* 842 (2020).
- [46] Y. Liu, L. Zhang, D. Liu, W. Hu, X. Yan, C. Yu, H. Zeng, T. Shen, Turbostratic carbon-localised FeS₂ nanocrystals as anodes for high-performance sodium-ion batteries, *Nanoscale* 11 (2019) 15497–15507.